A five-count reserve without a displacement error

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The crossing identity gives nonnegative local losses. We bound five of them directly, by replacing their kernels with a smooth function before comparing conditional sums. Centering makes the replacement cost proportional to the noise variance. There is no additive inward-displacement error and no assumption that the two conditional laws coincide.

Let U_0, U_1 be centered with

$$-3/5 \le U_0 \le a_0, \qquad -a_1 \le U_1 \le 3/5, \qquad a_0, a_1 \ge 0, \qquad a_0 + a_1 \le 9/125.$$

For $n \ge 42$ and $p \in [2/5, 53/100]$, write $q = 1 - p$, $s_j = \mathbb{E}U_j^2$, $v = pq$ and $\lambda = n(qs_0 + ps_1) \le 9/25$. All conditional sums below use independent coordinates. Set

$$h(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - 1 - U_1)_+ - \mathbb{P}(1 + U_1 \le x),$$

$$V_i = \sum_{l<i} U_{1l} + \sum_{l>i} U_{0l}, \qquad g_i(t) = \mathbb{E}h(t - i + 1 - V_i).$$

The retained crossing identity proves $g_i \ge 0$ and $\sum_i g_i = t - \mathbb{E}K$ whenever $na_0 \le t < n(1 - a_1)$.

Theorem 1 (Five-count reserve)**.** *Suppose $|t - np| < 47/50$, and put $k = \lfloor t \rfloor$, $a = 1 - 5/n$ and $b = 1 + 3/n$. Then*

$$\sum_{i=k-1}^{k+3} g_i(t) \ge \frac{2}{5}\left\{\frac{1 - e^{-3a\lambda}}{2} - \frac{(9/25)b\lambda + 3b^2\lambda^2}{16} - \frac{10\pi^2\lambda}{n}\right\}. \tag{1}$$

In particular the following uniform bounds hold:

n *at least*	125	150	200	250	300	400	500
$\sum_{i=k-1}^{k+3} g_i/\lambda$ *at least*	1/250	1/20	1/8	1/6	19/100	9/40	6/25.

At $\lambda = 0$ these statements mean the corresponding nonnegative bound, without division. The five indices lie in $\{1, \dots, n\}$.

For the acquired raw thresholds $t = np + H\sqrt{v + \lambda/n}$ with $|H| \le 46/25$, the required displacement follows already at $n = 42$:

$$(46/25)^2(1/4 + (9/25)/42) < (47/50)^2.$$

The result therefore covers every phase and every threshold atom in the right strip. It does not require the geometric budget or $d \le 3/n$. It is a lower bound for actual local losses, not a complete CDF comparison. For any nonnegative weights π_i, it immediately yields a weighted reserve by multiplying the right side of (1) by $\min_{k-1 \le i \le k+3} \pi_i$. No distant-count tail is then needed.

1 A compact minorant of each crossing kernel

Define the continuously differentiable function

$$B(x) = \begin{cases} \sin^2(\pi x), & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Its derivative is globally Lipschitz with constant $2\pi^2$: the derivative vanishes at both joins and its derivative in each open piece is bounded in absolute value by $2\pi^2$.

Lemma 2. *For every real x, one has $h(x) \ge (2/5)B(x)$.*

Proof. For $x < 1 - a_1$, $h(x) = \mathbb{E}(x - U_0)_+ \ge 0$. For $x \ge 1 - a_1 > a_0$, put–call parity gives $h(x) = \mathbb{P}(1 + U_1 > x) - \mathbb{E}(1 + U_1 - x)_+$. This is at least $(x - 3/5)\mathbb{P}(1 + U_1 > x) \ge 0$ until $x = 8/5$, and equals zero beyond that endpoint. Thus $h \ge 0$ globally. For $0 \le x < 1 - a_1$, the terms involving $1 + U_1$ vanish and Jensen gives $h(x) = \mathbb{E}(x - U_0)_+ \ge x$. We claim $B(x) \le (5/2)x$ on $[0, 1]$. For $x \le 1/4$, use $\sin^2(\pi x) \le \pi^2 x^2$ and $\pi^2 < 10$. For $x \ge 2/5$, use $B \le 1$. Between these points, concavity and the tangent at $1/4$ give

$$B(x) \le \tfrac{1}{2} + \pi(x - \tfrac{1}{4}) \le \tfrac{5}{2}x.$$

The last affine inequality holds at both endpoints, using $\pi < 10/3$.

For $1 - a_1 \le x \le 1$, put $D = 1 + U_1$ and $L = \mathbb{E}(D - x)_+$. Put–call parity gives $h(x) = \mathbb{P}(D > x) - L$, with a strict survival event. Since $D - x \le 1 + 3/5 - x$ on this event and $\mathbb{E}D = 1$,

$$h(x) \ge \frac{x - 3/5}{1 + 3/5 - x} L \ge \frac{41}{84}(1 - x).$$

Here $x \ge 1 - a_1 \ge 116/125$ and $L \ge 1 - x$. On the other hand

$$\frac{2}{5}B(x) \le \frac{2}{5}\pi^2(1 - x)^2 \le \frac{2}{5}\left(\frac{22}{7}\right)^2 \frac{9}{125}(1 - x) < \frac{41}{84}(1 - x).$$

At $x = 1$ the minorant is zero. This includes all atoms of D because the displayed survival event stays strict. □

2 Four centered replacements suffice

We use the following elementary smooth comparison. If F' is Lipschitz with constant L and Z is centered, then

$$|\mathbb{E}F(r + Z) - F(r)| \le \tfrac{L}{2}\mathbb{E}Z^2.$$

Taylor's integral remainder proves this even if F'' exists only piecewise. Conditional on an independent random r, it follows that replacing a centered summand of variance s_0 by one of variance s_1 changes the expectation by at most $L(s_0 + s_1)/2$.

Write $u = t - k \in [0, 1)$ and freeze the middle omitted-coordinate sum

$$V = V_{k+1} \overset{\text{law}}{=} U_{11} + \cdots + U_{1k} + U_{0,k+2} + \cdots + U_{0n}.$$

For $j = -2, -1, 0, 1, 2$, let $V^{(j)}$ have the law of V_{k+1+j}, and put $B_j(x) = B(u - j - x)$. All five sums have $n - 1$ terms. For $j > 0$, the law of $V^{(j)}$ replaces j copies of U_0 by U_1. The two expectation differences regroup exactly as

$$\begin{aligned} &\mathbb{E}B_1(V^{(1)}) + \mathbb{E}B_2(V^{(2)}) - \mathbb{E}B_1(V) - \mathbb{E}B_2(V) \\ &= \{\mathbb{E}(B_1 + B_2)(V^{(1)}) - \mathbb{E}(B_1 + B_2)(V)\} \\ &\qquad + \{\mathbb{E}B_2(V^{(2)}) - \mathbb{E}B_2(V^{(1)})\}. \end{aligned}$$

Each brace is one centered replacement. The two translated bumps in $B_1 + B_2$ have disjoint open supports, so its derivative still has Lipschitz constant $2\pi^2$, rather than twice that value. The negative indices have the identical regrouping, now replacing U_1 by U_0. The derivatives are zero at every common boundary. Consequently

$$\sum_{j=-2}^{2} \mathbb{E}B(u - j - V^{(j)}) \geq \sum_{j=-2}^{2} \mathbb{E}B(u - j - V) - 4\pi^2(s_0 + s_1). \tag{2}$$

There are enough copies of each law: $k - 1 \geq 1$ and $k + 3 \leq n$ follow from $n \geq 42$, the specified p range and the raw displacement bound. No conditional sum at one count is assumed independent of a sum at another.

The translates partition a periodic sine square:

$$\sum_{j\in\mathbb{Z}} B(u - j - z) = \sin^2(\pi(u - z)).$$

Every omitted nonzero term, with $|j| \geq 3$, requires $|z| > 2$. At integer joins both adjacent bumps vanish. Therefore

$$\sum_{j=-2}^{2} \mathbb{E}B(u - j - V) \geq \mathbb{E}\sin^2(\pi(u - V)) - \frac{\mathbb{E}V^4}{16}. \tag{3}$$

3 Variance and a single characteristic function

Put $\delta = k - np \in (-97/50, 47/50)$. The actual frozen variance is

$$\tau = \mathbb{E}V^2 = (nq - \delta - 1)s_0 + (np + \delta)s_1, \qquad a\lambda \leq \tau \leq b\lambda.$$

Indeed $p \geq 2/5$, $q \geq 47/100$, so the lower coefficient losses are at most $5/n$ and the upper gains at most $3/n$. Since $|U_j| \leq 3/5$, independence gives

$$\mathbb{E}V^4 \leq (9/25)\tau + 3\tau^2 \leq (9/25)b\lambda + 3b^2\lambda^2.$$

Also $s_0 + s_1 \leq 5\lambda/(2n)$.

For completeness, for $|x| \leq 84/125$,

$$1 - \cos(2\pi x) \geq 3x^2.$$

On $0 \leq x \leq 2/3$, concavity gives $\sin(\pi x) \geq (3\sqrt{3}/4)x > (51/40)x$. On $2/3 \leq x \leq 84/125$, the Lipschitz bound for sine gives $\sin(\pi x) > 17/20 - 44/2625$. Squaring these positive estimates proves the claim by rational arithmetic, using $\sqrt{3} > 17/10$ and $\pi < 22/7$.

For either conditional noise Z and its independent copy Z', $|Z - Z'| \leq 3/5 + 9/125 = 84/125$. It follows that

$$1 - |\mathbb{E}e^{2\pi iZ}|^2 = \mathbb{E}\{1 - \cos(2\pi(Z - Z'))\} \geq 6\operatorname{Var} Z.$$

Thus $|\mathbb{E}e^{2\pi iZ}| \leq e^{-3\operatorname{Var} Z}$. Independence of the $n - 1$ frozen summands, not a claim about the selected count, gives

$$\mathbb{E}\sin^2(\pi(u - V)) \geq \frac{1 - |\mathbb{E}e^{2\pi iV}|}{2} \geq \frac{1 - e^{-3\tau}}{2} \geq \frac{1 - e^{-3a\lambda}}{2}.$$

Combine this with Lemma 2, (2) and (3) to prove (1).

To verify the table without floating-point exponentials, put $L = 9/25$, $a_N = 1 - 5/N$, $b_N = 1 + 3/N$ and $P_6(x) = \sum_{j=0}^{6} x^j/j!$. Concavity of $1 - e^{-3a_N\lambda}$ on $[0, L]$ gives the rational lower coefficient

$$c_N = \frac{2}{5}\left\{\frac{1 - P_6(3a_NL)^{-1}}{2L} - \frac{(9/25)b_N + 3b_N^2L}{16} - \frac{10(22/7)^2}{N}\right\}.$$

Here $e^x \geq P_6(x) > 0$, and increasing n improves each term. Each c_N strictly exceeds its displayed table value by exact rational arithmetic. The argument is uniform down to zero noise.

Scope and dependencies

Only elementary inequalities and the local crossing kernel are used. The kernel's nonnegativity is also proved above by the same separated-support argument as in the retained v13 crossing note. Identifying the full sum with $t - \mathbb{E}K$, or joining these local losses to the actual CDF, uses that note's exact identities with their endpoint conventions. The five-count reserve needs no external probability theorem, finite certificate, acquisition argument, common conditional law, or selected-index independence. It does not itself settle the remaining normal, curvature and binomial-weight corrections. The table does not claim a positive bound for every $42 \le n < 125$.